Hein Zarni Aung

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EDUCATION

Stony Brook University Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Relevant Coursework: Systems Fundamentals I & II, Software Security, Full Stack Development

INTERNSHIPS

Software Engineering Intern — Clustr Jan 2026 – Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering.
- Implemented secure authentication features, including a password reset flow (client screen plus backend cloud function) and custom claims to drive new-user onboarding logic.

Software Engineering Intern — Prenora Dynamics LLC Jun 2026 – Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors, using the Supabase CLI to manage schema migrations and configure user auth/login flows.

PROJECTS

TuneAcademy Python, FastAPI, Docker, NumPy, Essentia, Firestore

- Set up a CI/CD pipeline using GitHub Actions to lint, test, and auto-build/deploy a Python/FastAPI backend running in Docker, with a Cloud Firestore database and GCP hosting, deployed to Cloudflare Workers.
- Built the AI/audio-analysis pipeline for an app matching music students with instructors, extracting pitch, rhythm, and tone features with Essentia/NumPy and pivoting to rule-based scoring algorithms under hackathon time and compute constraints.

Discord Voice Channel Activity Monitor Node.js, Discord.js, PM2, AWS

- Built a Node.js/Discord.js automation bot with JSON-based configuration for per-server/per-user channel thresholds and customizable notifications, deployed on a cloud server using PM2 and reaching 50 concurrent active users.

RESEARCH

Undergraduate Research Assistant — Ethos Lab, Stony Brook University Sep 2025 – Dec 2025

PI: Abisheka Pitumpe | Team: 2

- Built a pipeline using Python and Selenium to monitor newly issued SSL/TLS certificates in real time, scraping HTML/CSS/JS/URL content to flag domains impersonating EZ-Pass tolling services.
- Collaborated in a two-person team to design and validate keyword- and heuristic-based detection logic, manually triaging hundreds of flagged domains to distinguish real fraud from noise.

CAMPUS INVOLVEMENT

Teaching Assistant — Data Structures & Algorithms, Stony Brook University Aug 2025 – Dec 2025

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations; created original midterm review materials adopted as the basis for later semesters and hosted review sessions reaching 100+ students.

SKILLS

Languages: Python, JavaScript, TypeScript, Java, C

Backend: FastAPI, Node.js, Express.js

Databases & Cloud: PostgreSQL, Firebase/Firestore, MongoDB, Supabase, AWS, GCP

Tools: Git, Docker, Linux, PM2, Selenium

Frontend: React, React Native, Next.js